

factory operations are critical. These skills are also highly transferable. An industrial safety engineer trained in semiconductor manufacturing can work in the chemical sector or take on roles abroad.

What other types of units are emerging to support the semiconductor ecosystem in India?

Applied Materials and Lam Research have both announced major investments. Applied Materials began with a \$400 million project in India. They have already acquired a ₹3.5 billion land parcel in Bengaluru to build a supplier and test ecosystem. This is prompting their global supply chain partners to take India seriously. For instance, Rockwell Automation has announced a \$66 million project in Chennai to build intelligent power systems for Applied's global equipment. That is just one example; similar developments are also taking place with other companies.

Fabs also need speciality gases, bulk gases, ultra-pure water, sensors, and much more. Water technology becomes especially critical, precision in every drop leads to profit in every wafer. Even cleanroom creation and maintenance are specialised engineering functions.

Could you elaborate more on the engineering behind fabs? What exactly happens beneath the cleanroom?

Sure. The real engineering beauty of a fab lies in the sub-fab, which is located below the cleanroom. This is where utility feeds, gas pipelines, safety systems, and abatement systems are installed. It involves extremely detailed industrial engineering. Even before construction begins, industrial engineers must map out the piping, feeds, sensor placements, scrubbers, and more. Every element must be pre-designed

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to ensure safety and maximise yield. It goes far beyond using EDA tools; precision engineering is required to build the infrastructure itself.

How do you see India overcoming the challenge of creating new skill sets, updating curriculums, and exposing students to new equipment and tools?

Glad you asked. The Electronics Sector Skill Council and MeitY have already initiated key efforts. I have participated in committees involving both academia and industry leaders. **We began by breaking down semiconductor manufacturing into unit processes, which were then translated into specific job roles and mapped to the necessary skills.**

These were developed into detailed skill curricula and qualification packs, which were shared with the industry and submitted to the Ministry of Skill Development and the National Council for Vocational Training (NCVT). After rigorous vetting, they were approved as National Occupational Standards, making them eligible for training under government schemes.

Simultaneously, this content was shared with AICTE. **On 9 April 2024, AICTE officially included semiconductor manufacturing electives in technical curricula. These can now**

be adopted by institutes and integrated into academic programmes.

You mentioned that engineers from multiple disciplines, such as chemical and electrical, are needed. Are other Sector Skill Councils also working on this, or is it just the electronics sector?

The Electronics Sector Skill Council has taken the lead in semiconductor skilling—even for roles typically linked to other engineering branches. For example, we have developed a comprehensive industrial safety curriculum specifically tailored to the requirements of the semiconductor industry. It covers hazardous chemical handling, electrical safety, latent energy control, and emergency response procedures. This content is rigorous enough to be transferable across various process industries, not just the electronics sector. So, while the ESSCI is currently leading this initiative, other councils are welcome to adopt these standards. The content itself is inherently multidisciplinary.

That means the curriculum is not restricted to electronics engineers, right? Even students from other branches, like chemical or civil engineering, can pursue it.

Absolutely. That is exactly how we have designed it. Water technology, cleanroom technology, safety, and plasma control are all specialisations that students from various engineering streams can add to their base degrees. **We do not believe in creating over-specialised individuals. Instead, students begin with a solid foundation degree and then pursue modular certifications to build expertise.** For example, someone may start as a HAZCHEM Safety Technician, then progress to Specialist, and eventually become an Expert with experience and revalidation. We have also ensured these certifications are